Part II, Lecture 1: Credit Risk Management Map and a First Insight into Default Risk

By the end of this session you must know what credit risk is, how the credit risk management map splits losses into expected and unexpected, how EL = PD x LGD x EAD works in the default-mode approach, which financial statement line absorbs each leg (provisions vs equity capital), how DM and MTM modelling approaches differ and which of the six credit risks each VaR model type covers, and how rating systems measure PD through assignment (differentiation) and quantification (calibration) with long-run historical default rates.

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1. The Bank's Problem: Innovatech vs SolidSteel

The session opens with a lending choice. Innovatech, a promising technology company, wants a 1 million euro loan to launch a new product: high growth potential, high risk of failure. SolidSteel, an established manufacturer with stable but low profits, wants 1 million euros to modernize a plant: low risk, low return for the bank.

The point of the exercise: a bank's job is not to avoid risks, but to measure, price and manage them correctly. Both loans can be good business if the risk is quantified and priced.

2. What Is Credit Risk?

Credit risk: the possibility of a loss resulting from a borrower's or counterparty's failure to repay a loan or meet contractual obligations.

Managing it effectively is crucial for a bank's profitability and solvency.

3. Expected Loss vs Unexpected Loss: The Two Legs

Credit losses are viewed from two perspectives, the two legs of credit risk management:

Expected loss (EL): the average loss anticipated over a given period for a portfolio. Predictable and recurring in the long run.

Unexpected loss (UL): the potential variability of losses around the expected average. The risk of larger-than-average losses within a given time horizon, the tail risk.

The financial statement treatment differs by leg, and this is heavily tested:

Must know:
• Expected losses lead to **credit loss provisions**: recorded as an expense on the income statement, they reduce the net value of loans on the balance sheet via an allowance account. EL is a cost of doing business
• Unexpected losses lead to **equity capital requirements**: capital absorbs losses beyond provisions and ensures the bank's solvency. Equity holders bear this risk

All six credit risks (default, severity, exposure, migration, spread, concentration) can be viewed from both the expected and the unexpected perspective, except concentration risk: concentration is inherently a portfolio effect (how an exposure correlates with others) and belongs to the unexpected, capital side.

4. Measuring EL: PD, LGD, EAD

In the default-mode (DM) approach, which assumes losses only derive from default events, expected loss needs three measures:

PD (Probability of Default): the likelihood that a borrower defaults within a specific time horizon, for example one year. Between 0% and 100%.

LGD (Loss Given Default): the proportion of the exposure lost if a default occurs, after accounting for recovery. Between 0% and 100%.

EAD (Exposure at Default): the outstanding amount of the credit exposure at the time default occurs. A monetary value.

Core formula: EL = PD x LGD x EAD

5. Measuring UL: VaR, MRC and the Model Families

Unexpected loss is measured with:

- **VaR (Value at Risk):** the maximum potential loss over a given time horizon at a specific confidence level, for example 99.9% VaR over one year.
- **MRC (Minimum Required Capital):** regulatory capital set by the Basel frameworks (I, II, III and following), based on risk-weighted assets and incorporating unexpected loss concepts.

And managed with two model families plus regulation:

- **Simplified Credit Risk models (SCR):** focus on stand-alone risk.
- **Full Portfolio Credit Risk models (FPCR):** include portfolio effects, concentration and correlation.
- **Regulatory rules (Basel):** provide methods and parameters for calculating MRC.

Historical anchor: JP Morgan's CreditMetrics, developed in the late 1990s, was the first major MTM full-portfolio model. It marked the shift towards portfolio-wide risk measurement including migration and spread effects, and it is still used by sophisticated banks for complex portfolios.

6. Default-Mode vs Mark-to-Market

Default-Mode (DM): a credit loss only occurs when a borrower officially defaults.

Mark-to-Market (MTM): a credit loss can occur from any decline in the market value of the exposure, even without default, for example after a rating downgrade or spread widening.

Which risks each model type covers:

Must know:
• DM models (typically SCR): default risk (PD), severity risk (LGD), exposure risk (EAD)
• DM full-portfolio (FPCR) models: those three plus concentration risk
• MTM full-portfolio models: all six risks, adding migration risk and spread risk

7. Rating Systems and PD Estimation

To measure PD, lenders use rating systems that produce borrower or issuer ratings. As S&P says, a rating is, in the end, an opinion. Ratings are categorical ordinal measures of risk (AAA, AA, …, CCC, D).

Two distinct processes, easy to confuse:

- **Rating assignment (risk differentiation):** assigning a borrower to a rating class from factors like financials, industry, management. The output is a graded opinion, a rank-ordered classification.
- **Rating quantification (calibration):** associating a cardinal PD (a number between 0 and 1) to each rating class, typically over a one-year horizon.

The most common quantification method is empirical: PD is estimated as the long-run historical average of one-year default rates for each rating class. This is also the regulatory approach: Basel II (paragraphs 447, 461, 463) requires PD estimates to be a long-run average of one-year default rates, with at least five years of underlying observations for at least one source. The ECB's 2017 Targeted Review of Internal Models specifies the default rate computation: the denominator is the number of non-defaulted obligors with any credit obligation at the start of the one-year period, the numerator those with at least one default event during the year.

The Parmalat case frames the scepticism: the Italian dairy multinational collapsed in December 2003 with a 14 billion euro hole, Europe's biggest bankruptcy, while rated BBB- by S&P until a few days before. Ratings fail; the question is whether they fail on average or systematically.

8. Common Mistakes and Exam Traps

- Mixing up the financial statement treatment: provisions cover expected losses, equity capital covers unexpected losses. Reversing the two is the classic error.
- Believing all six credit risks have both an expected and an unexpected side. Concentration risk is the exception: it only exists at the unexpected, portfolio level.
- Confusing assignment with quantification. Assignment produces the ordinal class (the letter), quantification attaches the cardinal PD (the number).
- Treating EL as a risk buffer. EL is a predictable cost priced into the business and provisioned; capital is the buffer for what exceeds expectations.

- Assuming a default is needed for a credit loss under every approach. Only DM works that way; MTM recognises losses from downgrades or spread widening without any default.

9. Must Know Absolutely

Must know:
• Credit risk = possibility of loss from a borrower failing to meet obligations. Banks measure, price and manage it, they do not avoid it
• EL = PD x LGD x EAD, covered by provisions through the income statement. UL is the volatility around EL, covered by equity capital
• Concentration risk is the only one of the six risks with no expected-loss side
• DM models cover PD, LGD, EAD; DM portfolio models add concentration; MTM portfolio models cover all six risks. CreditMetrics (JP Morgan, late 1990s) was the first major MTM portfolio model
• Rating = ordinal opinion from assignment; PD = cardinal calibration, estimated as the long-run historical average of one-year default rates per class, with at least five years of data per Basel II